

Notes

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Transform original event of Taylor polynomials by applying a Markov inequality to obtain an upper bound dependent on the moment generating function of our random variable. By independence of the random variables, we can write the expectation of the exponential of the sum as a product of each individual expectation. Then we can optimize this final bound in terms of λ .

Bernoulli Sums:

$$\begin{aligned} X_1, \dots, X_n &\text{ iid Bern}(p) \\ E[X_i] &= p, \text{Var}(X_i) = p(1-p) \\ P[X_i = 1] &= p, P[X_i = 0] = 1-p \\ \mathcal{N}(0, 1) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - E[X_i]}{\sqrt{\text{Var}(X_i)}} \\ P\left[\frac{1}{n} \sum_{i=1}^n X_i - p \geq t\right] &= P\left[\uparrow \geq t \sqrt{\frac{n}{p(1-p)}}\right] \approx 1 - \Phi\left(t \sqrt{\frac{n}{p(1-p)}}\right) \approx e^{-\frac{nt^2}{p(1-p)}} \end{aligned}$$

Large Deviation Principle:

$$\begin{aligned} P\left[\frac{1}{n} \sum_{i=1}^n X_i - p \geq t\right] &\approx e^{-nD_{KL}(p+t||p)} \\ D_{KL}(q||p) &= E_q\left[\log\left(\frac{q}{p}\right)\right] = (1-q) \log\left(\frac{1-q}{1-p}\right) + q \log\left(\frac{q}{p}\right) \approx \frac{(q-p)^2}{2p(1-p)}, q = t \end{aligned}$$

by Chebyshev inequality:

$$\begin{aligned}
P\left[\frac{1}{n} \sum_{i=1}^n X_i - p \geq t\right] &\leq P\left[\left(\frac{1}{n} \sum_{i=1}^n X_i - p\right)^2 \geq t^2\right] \\
&\leq \frac{1}{t^2} E\left[\left(\frac{1}{n} \sum_{i=1}^n X_i - p\right)^2\right] \\
&= \frac{1}{t^2} \frac{p(1-p)}{n}
\end{aligned}$$

in the case of polynomial decay k :

$$\begin{aligned}
P\left[\frac{1}{n} \sum_{i=1}^n X_i - p \geq t\right] &\leq P\left[\left(\frac{1}{n} \sum_{i=1}^n X_i - p\right)^k \geq t^k\right] \\
&\leq \frac{1}{t^k} E\left[\left(\frac{1}{n} \sum_{i=1}^n X_i - p\right)^k\right] \\
&= \left(\frac{1}{t^2} \frac{p(1-p)}{n}\right)^{\frac{k}{2}}
\end{aligned}$$

Let $\psi(\lambda) = \log E[e^{\lambda(X_i-p)}]$ where it's MGF is $e^{\psi(\lambda)} = E[e^{\lambda(X_i-p)}]$, thus:

$$\begin{aligned}
P\left[\frac{1}{n} \sum_{i=1}^n X_i - p \geq t\right] &\leq P[e^{\lambda \sum_{i=1}^n (X_i-p)} \geq e^{\lambda nt}], \lambda > 0 \\
&\leq e^{-\lambda nt} E[e^{\lambda \sum_{i=1}^n (X_i-p)}] \\
&= e^{-\lambda nt} \prod_{i=1}^n E[e^{\lambda (X_i-p)}] \\
&= e^{-\lambda nt + n\psi(\lambda)} \\
&= e^{-n(\lambda t - \psi(\lambda))}
\end{aligned}$$

Maximizing $\lambda t - \psi(\lambda)$ gives the best bound (minimizes due to $e^{-n(\dots)}$)

Given $\psi(\lambda) = \log E[e^{\lambda X_i}] - \lambda p = \log(1 - p + pe^\lambda) - \lambda p$:

$$\begin{aligned}
\lambda t - \psi(\lambda) &= \lambda t - \log(1 - p + pe^\lambda) + \lambda p \\
&= \lambda(t + p) - \log(1 - p + pe^\lambda), \text{ minimize w/ derivative} = 0 \\
&= D_{KL}(p + t || p)
\end{aligned}$$

Thus the bound is $P[\frac{1}{n} \sum_{i=1}^n X_i - p \geq t] \leq e^{-nD_{KL}(p+t||p)}$. With this we can characterize how fast the probability decays as n increases.

This gives us the Chernoff bound, where for i.i.d. $X_i, \dots, X_n$ where $E[X_i] = 0$, $\psi(\lambda) = \log E[e^{\lambda X_i}]$:

$$\psi^*(t) = \sup_{\lambda > 0} (\lambda t - \psi(\lambda))$$

Then $P[\frac{1}{n} \sum_{i=1}^n X_i \geq t] \leq e^{-n\psi^*(t)}$.

$$X_i \sim \mathcal{N}(0, \sigma^2), \psi(\lambda) = \frac{\sigma^2 \lambda^2}{2}$$

$$\sup_{\lambda > 0} (\lambda t - \frac{\sigma^2 \lambda^2}{2})$$